

Algebra II with Trigonometry

Chapter 6.5

Olympiad Problems

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[4533652_alg-2trig-h-chp-6-5-note.pdf](#)

Problems

1. Evaluate $\cos^{-1}(\sin(\sin^{-1} \frac{3}{5}))$ without a calculator, expressing your answer in radians.
2. If $\tan^{-1}(x) + \tan^{-1}(y) = \frac{\pi}{4}$, what is the value of $x + y + xy$?
3. Simplify $\sin(\cos^{-1}(x)) + \cos(\sin^{-1}(x))$ for $x \in [-1, 1]$.
4. Without evaluating directly, decide if the following two are equal for $x \in [-\frac{\pi}{2}, \frac{\pi}{2}]$: $\arccos(\sin x)$ and $\arcsin(\cos x)$.
5. Find the exact value of $\tan(\arccos(\frac{1}{3}))$ in simplest radical form.
6. Let $A = \arctan 2 + \arctan 3$. Find the exact value of $\tan A$.